EE4047 – High Voltage Techniques I

Practices

Week 7.2 – Tuesday Lesson

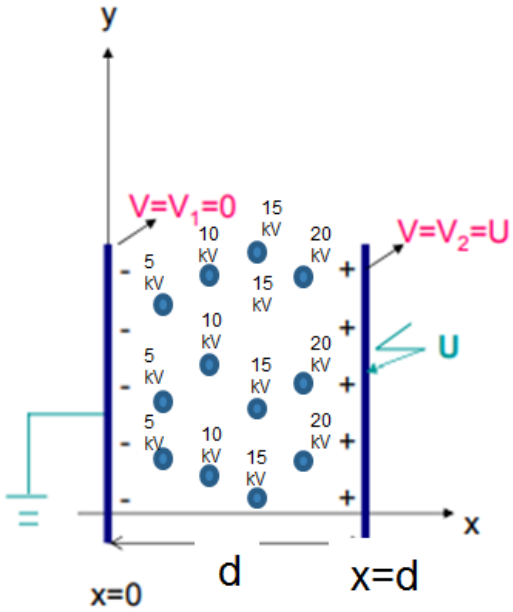
Asst. Prof. Dr. M. Alparslan Zehir

05.04.2022

1 EXAMPLE 1

Example-1: For the infinite length planar electrode system shown in the figure, U=25 kV (rms, effective) and d=1 cm. The potential values measured at several points of the dielectric material are provided in the figure.

a) Draw the equipotential lines and field lines.
b) Calculate the electric field strength.
c) For the dielectric breakdown strength $E_b = 30\ kV/cm$, find out if the system will withstand or breakdown.



1 EXAMPLE 1

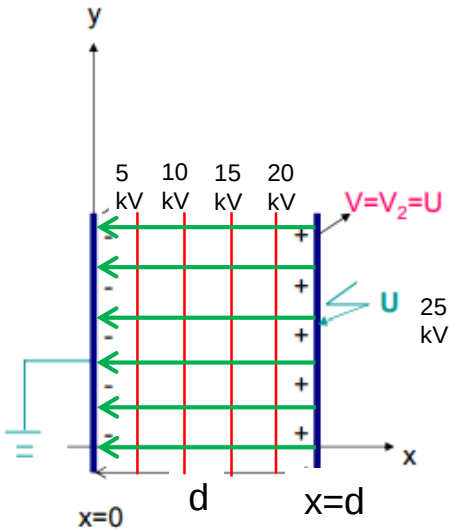
Example-1: For the infinite length planar electrode system shown in the figure, U=25 kV (rms, effective) and d=1 cm.

a) Draw the 20 kV, 15 kV, 10 kV and 5kV equipotential lines and draw field lines.

b) Calculate the electric field strength.

c) For the dielectric breakdown strength $E_b = 30\ kV/cm$, find out if the system will witstand or breakdown.

Solution-1: a)



1 EXAMPLE 1

Example-1: For the infinite length planar electrode system shown in the figure, U=25 kV (rms, effective) and d=1 cm.

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b) Calculate the electric field strength.
c) For the dielectric breakdown strength $E_b = 30\ kV/cm$, find out if the system will withstand or breakdown.

Solution-1: b) The peak value of the applied voltage: $U_{peak} = \sqrt{2}.25 = 35{,}35\ kV$

For planar electrode system $E = E_{max} = E_{min} = \frac{U}{d} = \frac{35{,}35}{1} = 35{,}35\ kV/cm$

c) $E_{max} = 35{,}35\ kV/cm > E_b = 30\ kV/cm$ the system cannot withstand to the applied voltage, there is breakdown

EXAMPLE 2

Example-2: In the multilayer planar electrode system given in the figure, calculate the maximum voltage which will not cause breakdown at any layer.

Solution-2: Electric field strength formula for a multilayer dielectric planar electrode system:

$$E = \frac{U}{\varepsilon_i A} \rightarrow E_{d_i} = \frac{U_d}{\varepsilon_i A} \rightarrow U_d = E_{d_i}\varepsilon_i A$$

$$A = \sum_{i=1}^{3} \frac{d_i}{\varepsilon_i} = \frac{1}{\varepsilon_0}\left(\frac{2}{4} + \frac{3}{8} + \frac{5}{8}\right) = \frac{1}{\varepsilon_0}.\frac{4+3+5}{8} = \frac{1}{\varepsilon_0}.\frac{3}{2}$$

For layer 1:

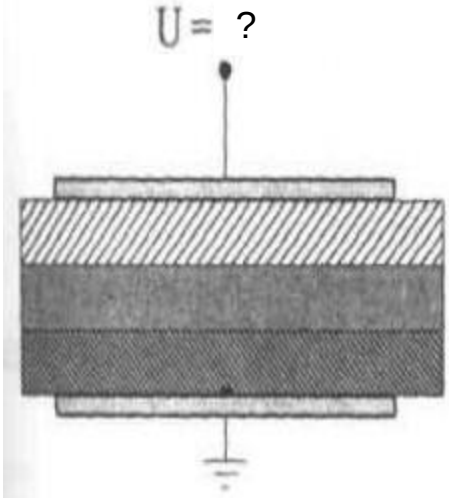
$$U_d = E_{d_1}\varepsilon_1 A = 90.4.\varepsilon_0.\frac{3}{\varepsilon_0.2} = 540\ kV$$

For layer 2:

$$U_d = E_{d_2}\varepsilon_2 A = 45.8.\varepsilon_0.\frac{3}{\varepsilon_0.2} = 540\ kV$$

For layer 3:

$$U_d = E_{d_3}\varepsilon_3 A = 30.8.\varepsilon_0.\frac{3}{\varepsilon_0.2} = 360\ kV$$



ε_{r1}=4 d1=2cm E_{d1} = 90 kV/cm

ε_{r2}=8 d2=3cm E_{d2} = 45 kV/cm

ε_{r3}=8 d3=5cm E_{d3} = 30 kV/cm

1 EXAMPLE 3

Example-3: In the multilayer concentric spherical electrode system with two dielectric layers shown in the figure below, $U = 120\ kV$, $r_1 = 0{,}8\ cm$, $r_2 = 3{,}4\ cm$, $r_3 = 4{,}6\ cm$, $\varepsilon_{r1} = 3{,}8$ and $\varepsilon_{r2} = 2{,}5$.

a) Calculate the potential, minimum and maximum electric fields at each layer

b) Considering breakdown strength for the layers as $E_{b1} = 150\ kV/cm$ and $E_{b2} = 130\ kV/cm$ find out if the system will withstand or not.

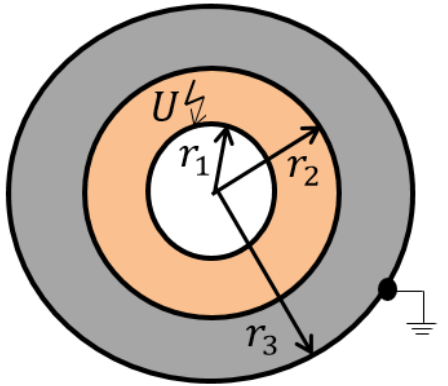
a)

$$A = \frac{1}{3{,}8}\frac{3{,}4-0{,}8}{0{,}8.3{,}4} + \frac{1}{2{,}5}\frac{4{,}6-3{,}4}{3{,}4.4{,}6} = 0.28$$

$$U_1 = \frac{120}{0{,}28.3{,}8}\frac{3{,}4-0{,}8}{3{,}4.0{,}8} = 107.81\ kV$$

$$E_{1-max} = 107.81\frac{0{,}8.3{,}4}{(3{,}4-0{,}8)}\frac{1}{0{,}8^2} = 176.23\ kV/cm$$

$$E_{1-min} = 107.81\frac{0{,}8.3{,}4}{(3{,}4-0{,}8)}\frac{1}{3{,}4^2} = 9.76\ kV/cm$$



1 EXAMPLE 3

Example-3: In the multilayer concentric spherical electrode system with two dielectric layers shown in the figure below, $U = 120\ kV$, $r_1 = 0{,}8\ cm$, $r_2 = 3{,}4\ cm$, $r_3 = 4{,}6\ cm$, $\varepsilon_{r1} = 3{,}8$ and $\varepsilon_{r2} = 2{,}5$.

a) Calculate the potential, minimum and maximum electric fields at each layer

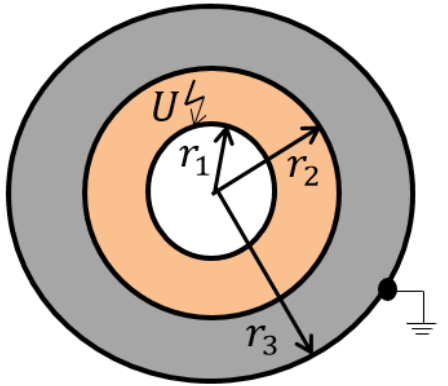
b) Considering breakdown strength for the layers as $E_{b1} = 150\ kV/cm$ and $E_{b2} = 130\ kV/cm$ find out if the system will withstand or not.

a)

$$U_2 = \frac{120}{0{,}28.2{,}5}\frac{4{,}6 - 3{,}4}{4{,}6.3{,}4} = 13.15\ kV$$

$$E_{2-max} = 13.15\frac{3{,}4.4{,}6}{(4{,}6 - 3{,}4)}\frac{1}{3{,}4^2} = 14.83\ kV/cm$$

$$E_{2-min} = 13.15\frac{3{,}4.4{,}6}{(4{,}6 - 3{,}4)}\frac{1}{4{,}6^2} = 8.10\ kV/cm$$



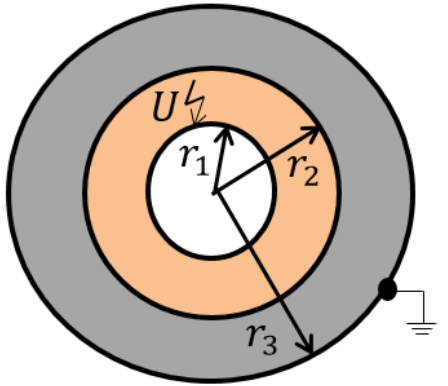
EXAMPLE 3

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a) Calculate the potential, minimum and maximum electric fields at each layer

b) Considering breakdown strength for the layers as $E_{b1} = 150\ kV/cm$ and $E_{b2} = 130\ kV/cm$ find out if the system will withstand or not.

b)



$E_{1-max} = 176.23 \frac{kV}{cm} > E_{b1} = 150\ kV/cm$ layer 1 cannot withstand, it breaksdown

$E_{2-max} = 14.83 \frac{kV}{cm} < E_{b2} = 130\ kV/cm$ layer 2 withstands

The new voltage on layer 2: $U_{2-new} = 120\ kV$

$E_{2-max-new} = 120 \frac{3{,}4.4{,}6}{(4{,}6-3{,}4)} \frac{1}{3{,}4^2} = 135.29\ kV/cm$ layer 2 cannot withstand, it breaksdo

system breaksdown.

END OF THE LESSON

Any questions?

Check out our playlist s on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ